

null hypothesis is true (**Held and Ott, 2018**). If the null hypothesis states the absence of an effect, a Bayes factor greater than one ($BF_{01} > 1$) indicates evidence for the absence of the effect and a Bayes factor smaller than one indicates evidence for the presence of the effect ($BF_{01} < 1$), whereas a Bayes factor not much different from one indicates absence of evidence for either hypothesis ($BF_{01} \approx 1$). Bayes factors are quantitative summaries of the evidence provided by the data in favor of the null hypothesis as opposed to the alternative hypothesis. If a dichotomous criterion for successful replication of a null result is desired, it seems natural to require a Bayes factor larger than some level $\gamma > 1$ from both studies, for example, $\gamma = 3$ or $\gamma = 10$ which are conventional levels for “substantial” and “strong” evidence, respectively (**Jeffreys, 1961**). In contrast to the non-significance criterion, this criterion provides a genuine measure of evidence that can distinguish absence of evidence from evidence of absence.

The main challenge with Bayes factors is the specification of the effect under the alternative hypothesis H_1 . The assumed effect under H_1 is directly related to the Bayes factor, and researchers who assume different effects will end up with different Bayes factors. Instead of specifying a single effect, one therefore typically specifies a “prior distribution” of plausible effects. Importantly, the prior distribution, like the equivalence margin, should be determined by researchers with subject knowledge and before the data are collected.

To compute the Bayes factors for the RPCB null results, we used the observed effect estimates as the data and assumed a normal sampling distribution for them (**Dienes, 2014**), as typically done in a meta-analysis. The Bayes factors BF_{01} shown in Figure 3 then quantify the evidence for the null hypothesis of no effect against the alternative hypothesis that there is an effect using a normal “unit-information” prior distribution (**Kass and Wasserman, 1995**) for the effect size under the alternative H_1 , see Appendix C for further details on the calculation of these Bayes factors. We see that in most cases there is no substantial evidence for either the absence or the presence of an effect, as with the equivalence tests. For instance, with a lenient Bayes factor threshold of 3, only 1 of the 15 replications are successful, in the sense of having $BF_{01} > 3$ in both the original and the replication study. The Bayes factors for the two previously discussed examples are consistent with our intuitions – in the **Goetz et al. (2011)** example there is indeed substantial evidence for the absence of an effect ($BF_{01} = 5$ in the original study and $BF_{01} = 4.1$ in the replication), while in the **Dawson et al. (2011)** example there is even anecdotal evidence for the *presence* of an effect, though the Bayes factors are very close to one due to the small sample sizes ($BF_{01} = 1/1.1$ in the original study and $BF_{01} = 1/1.8$ in the replication).

As with the equivalence margin, the choice of the prior distribution for the SMD under the alternative H_1 is debatable. The normal unit-information prior seems to be a reasonable default choice, as it implies that small to large effects are plausible under the alternative, but other normal priors with smaller/larger standard deviations could have been considered to make the test more sensitive to smaller/larger true effect sizes. The sensitivity analysis in Appendix A therefore also includes an analysis on the effect of varying prior standard deviations and the Bayes factor thresholds. However, again, to achieve replication success for a larger proportion of replications than the observed $1/15 = 7\%$, unreasonably large prior standard deviations have to be specified.

Of note, among the 15 RPCB null results, there are three interesting cases (the three effects from original paper 48 by **Lin et al., 2012** and its replication by **Lewis et al., 2018**) where the Bayes factor is qualitatively different from the equivalence test, revealing a fundamental difference between the two approaches. The Bayes factor is concerned with testing whether the effect is *exactly zero*, whereas the equivalence test is concerned with whether the effect is within an *interval around zero*. Due to the very large sample size in the original study ($n = 514$) and the replication ($n = 1'153$), the data are incompatible with an exactly zero effect, but compatible with effects within the equivalence range. Apart from this example, however, both approaches lead to the same qualitative conclusion – most RPCB null results are highly ambiguous.

Appendix B also shows Bayes factor analyses for the four study pairs with original null results from the RPP and EPRP. In contrast to the RPCB results, most Bayes factors indicate non-anecdotal

evidence for a null effect in cases where the non-significance criterion was met, possibly because of the larger sample sizes and smaller effects in these fields.

Box 1: Recommendations for the analysis of replication studies of original null results. Calculations are based on effect estimates $\hat{\theta}_i$ with standard errors σ_i from an original study ($i = o$) and its replication ($i = r$). Both effect estimates are assumed to be normally distributed around the true effect size θ with known variance σ_i^2 . The effect size θ_0 represents the value of no effect, typically $\theta_0 = 0$.

Equivalence test

1. Specify a margin $\Delta > 0$ that defines an equivalence range $[\theta_0 - \Delta, \theta_0 + \Delta]$ in which effects are considered absent for practical purposes.
2. Compute the TOST p -values for original ($i = o$) and replication ($i = r$) data

$$p_{\text{TOST},i} = \max \left\{ \Phi \left(\frac{\hat{\theta}_i - \theta_0 - \Delta}{\sigma_i} \right), 1 - \Phi \left(\frac{\hat{\theta}_i - \theta_0 + \Delta}{\sigma_i} \right) \right\},$$

with $\Phi(\cdot)$ the cumulative distribution function of the standard normal distribution.

```
## R function to compute TOST p-value based on effect estimate, standard error,
## null value (default is 0), and equivalence margin specified in step 1.
pTOST <- function(estimate, se, null = 0, margin) {
  p1 <- pnorm(q = (estimate - null - margin) / se)
  p2 <- 1 - pnorm(q = (estimate - null + margin) / se)
  p <- pmax(p1, p2)
  return(p)
}
```

3. Declare replication success at level α if $p_{\text{TOST},o} \leq \alpha$ and $p_{\text{TOST},r} \leq \alpha$, conventionally $\alpha = 0.05$.
4. Perform a sensitivity analysis with respect to the margin Δ . For example, visualize the TOST p -values for different margins to assess the robustness of the conclusions.

Bayes factor

1. Specify a prior distribution for the effect size θ that represents plausible values under the alternative hypothesis that there is an effect ($H_1 : \theta \neq \theta_0$). For example, specify the mean m and standard deviation s of a normal distribution $\theta | H_1 \sim N(m, s^2)$.
2. Compute the Bayes factors contrasting $H_0 : \theta = \theta_0$ to $H_1 : \theta \neq \theta_0$ for original ($i = o$) and replication ($i = r$) data. Assuming a normal prior distribution, the Bayes factor is

$$\text{BF}_{01,i} = \sqrt{1 + \frac{s^2}{\sigma_i^2}} \exp \left[-\frac{1}{2} \left\{ \frac{(\hat{\theta}_i - \theta_0)^2}{\sigma_i^2} - \frac{(\hat{\theta}_i - m)^2}{\sigma_i^2 + s^2} \right\} \right].$$

```
## R function to compute Bayes factor based on effect estimate, standard error,
## null value (default is 0), prior mean (default is null value), and prior
## standard deviation specified in step 1.
BF01 <- function(estimate, se, null = 0, priormean = null, priorsd) {
  bf <- sqrt(1 + priorsd^2/se^2) * exp(-0.5 * ((estimate - null)^2 / se^2 -
    (estimate - priormean)^2 / (se^2 + priorsd^2)))
  return(bf)
}
```

3. Declare replication success at level $\gamma > 1$ if $\text{BF}_{01,o} \geq \gamma$ and $\text{BF}_{01,r} \geq \gamma$, conventionally $\gamma = 3$ (substantial evidence) or $\gamma = 10$ (strong evidence).
4. Perform a sensitivity analysis with respect to the prior distribution. For example, visualize the Bayes factors for different prior standard deviations to assess the robustness of the conclusions.

Conclusions

The concept of “replication success” is inherently multifaceted. Reducing it to a single criterion seems to be an oversimplification. Nevertheless, we believe that the “non-significance” criterion – declaring a replication as successful if both the original and the replication study produce non-significant results – is not fit for purpose. This criterion does not ensure that both studies provide evidence for the absence of an effect, it can be easily achieved for any outcome if the studies have sufficiently small sample sizes, and it does not control the relevant error rates. While it is important to replicate original studies with null results, we believe that they should be analyzed using more informative approaches. Box 1 summarizes our recommendations.

Our reanalysis of the RPCB studies with original null results showed that for most studies that meet the non-significance criterion, the conclusions are much more ambiguous – both with frequentist and Bayesian analyses. While the exact success rate depends on the equivalence margin and the prior distribution, our sensitivity analyses show that even with unrealistically liberal choices, the success rate remains below 40% which is substantially lower than the 73% success rate based on the non-significance criterion.

This is not unexpected, as a study typically requires larger sample sizes to detect the absence of an effect than to detect its presence (*Matthews, 2006*, Section 11.5.3). Of note, the RPCB sample sizes were chosen so that each replication had at least 80% power to detect the original effect estimate based on a standard superiority test. However, the design of replication studies should ideally align with the planned analysis (*Anderson and Kelley, 2022*) so if the goal of the study is to find evidence for the absence of an effect, the replication sample size should be determined based on a test for equivalence, see *Flight and Julious (2015)* and *Pawel et al. (2023)* for frequentist and Bayesian approaches, respectively.

Our reanalysis of the RPP and EPRP studies with original null results showed that Bayes factors indeed indicate some evidence for no effect in cases where the non-significance criterion was satisfied, possibly due to the smaller effects and typically larger sample sizes in these fields compared to cancer biology. On the other hand, in most cases the precision of the effect estimates was still limited so that only one study pair achieved replication success with the equivalence testing approach. However, it is important to note that the conclusions from the RPP and EPRP analyses are merely anecdotal, as there were only four study pairs with original null results to analyze.

For both the equivalence test and the Bayes factor approach, it is critical that the equivalence margin and the prior distribution are specified independently of the data, ideally before the original and replication studies are conducted. Typically, however, the original studies were designed to find evidence for the presence of an effect, and the goal of replicating the “null result” was formulated only after failure to do so. It is therefore important that margins and prior distributions are motivated from historical data and/or field conventions (*Campbell and Gustafson, 2021*), and that sensitivity analyses regarding their choice are reported.

In addition, when analyzing a single pair of original and replication studies, we recommend interpreting Bayes factors and TOST p -values as quantitative measures of evidence and discourage dichotomizing them into “success” or “failure”. For example, two TOST p -values $p_{\text{TOST}} = 0.049$ and $p_{\text{TOST}} = 0.051$ carry similar evidential weight regardless of one being slightly smaller and the other being slightly larger than 0.05. On the other hand, when more than one pair of original and replication studies are analyzed, dichotomization may be required for computing an overall success rate. In this case, the rate may be computed for different thresholds that correspond to qualitatively different levels of evidence (e.g., 1, 3, and 10 for Bayes factors, or 0.05, 0.01, and 0.005 for p -values).

Researchers may also ask whether the equivalence test or the Bayes factor is “better”. We believe that this is the wrong question to ask, because both methods address different questions and are better in different senses; the equivalence test is calibrated to have certain frequentist error rates, which the Bayes factor is not. The Bayes factor, on the other hand, seems to be a